

Technology #159

Apply a covalently bonded heparin coating to
the luminal and external surfaces of the
tunneled hemodialysis catheter during
manufacturing

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply a covalently bonded heparin coating to the luminal and external surfaces of the tunneled hemodialysis catheter during manufacturing

WHY DOES IT MATTER?

Documented problem 'catheter thrombosis' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Hemostasis at the edge between physiology and cancer.'; 'Fibrinogen in extracellular matrix remodeling: functional switching, source heterogeneity, and biomaterial translation.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct a prospective, double-blind randomized controlled trial comparing heparin-coated versus standard catheters, measuring flow rates.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Antithrombotic heparin coating on the catheter surface that inhibits fibrin formation and platelet adhesion

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'catheter thrombosis' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Hemostasis at the edge between physiology and cancer.'; 'Fibrinogen in extracellular matrix remodeling: functional switching, source heterogeneity, and biomaterial translation.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Apply a covalently bonded heparin coating to the luminal and external surfaces of the tunneled hemodialysis catheter during manufacturing

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply a covalently bonded heparin coating to the luminal and external surfaces of the tunneled hemodialysis catheter during manufacturing

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * dP) / (8 * \mu * L)$.

Applicability to this invention: CONDITIONAL; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: NOT ESTABLISHED.

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * dP / \rho}$.

Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.
- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).

- Model output variables produced by the governing model: Q (m^3/s), Re (-), Q (m^3/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r, m] value_status=UNKNOWN; lumen length [r, m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP, Pa] value_status=UNKNOWN; flow rate range [Q, m^3/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'catheter thrombosis', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain at least 90 percent flow patency for at least	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Antithrombotic heparin coating on the catheter surface that inhibits fibrin form	MODELED
DI-003	Expected effect target	Sustained >90% flow patency for weeks to months, with a marked reduction	MODELED
DI-004	Domain design input pattern: required flow applicability	Variable expression [read]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: mechanism applicability	NOT_ESTABLISHED (candidate pattern; confirm against	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anticipated applicability	NOT_ESTABLISHED (candidate pattern; confirm against	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability	NOT_ESTABLISHED (candidate pattern; confirm against	ENGINEERING_PROPOSED
DI-008	Regulatory design input: compliance candidate standards	ISO 13485 (QMS) and further, pathway to EN 13061-1 (PCE) is never inferred	ENGINEERING_PROPOSED
DI-009	Custodied observation: Complications of tunneled hemodialysis	https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/ element at https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/	ENGINEERING_PROPOSED
DI-010	Custodied observation: Vascular access of tunneled hemodialysis	https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/ element at https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/	ENGINEERING_PROPOSED
DI-011	Custodied observation: Risk prediction of tunneled hemodialysis	https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/ element at https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/	ENGINEERING_PROPOSED
DI-012	Custodied observation: Heparin-coated tunneled hemodialysis	https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/ element at https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/	ENGINEERING_PROPOSED
DI-013	Custodied observation: Comprehensive management of tunneled hemodialysis	https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/ element at https://pubmed.ncbi.nlm.nih.gov/abstract/42857461/	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Apply a covalently bonded heparin coating	CONCEPTUAL	detailed architectural definition
DO-002	pressure regulation element realizing Apply a covalently bonded heparin coating	CONCEPTUAL	detailed architectural definition
DO-003	sensing / feedback element (if closed-loop) realizing Apply a covalently bonded heparin coating	CONCEPTUAL	detailed architectural definition
DO-004	termination interfaces (patient / reservoir) realizing Apply a covalently bonded heparin coating	CONCEPTUAL	detailed architectural definition
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal	CONCEPTUAL	interface materialization (if for application) bonded heparin coating
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop)	CONCEPTUAL	interface materialization (if for application) bonded heparin coating
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces	CONCEPTUAL	interface materialization (if for application) bonded heparin coating
DO-008	Device boundaries: patient/fluid inlet and outlet termination control/data processing	CONCEPTUAL	interface materialization (if for application) bonded heparin coating
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	covalently bonded heparin coating to the lumen
DO-010	Operating range for lumen inner diameter within which the mechanism holds its	UNKNOWN	stated behavior
DO-011	Operating range for lumen length within which the mechanism holds its stated	UNKNOWN	behavior or computed bounds for lumen length; verification of range endpoints
DO-012	Operating range for surface roughness within which the mechanism holds its	UNKNOWN	stated behavior
DO-013	Operating range for operating pressure head within which the mechanism holds	UNKNOWN	its stated behavior
DO-014	Operating range for flow rate range within which the mechanism holds its	UNKNOWN	stated behavior
DO-015	Operating range for viscosity of flowed medium within which the mechanism	UNKNOWN	holds its stated behavior
DO-016	No closed-loop control is proposed: the invention operates as open-loop as	CONCEPTUAL	specified; this record exists so the omission is explicit and auditable
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention
DO-018	Fixture enabling: bench flow-loop characterization across operating range	UNKNOWN	rig construction, build specifications, acceptance instrumentation
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying transitions)	UNKNOWN	fixture geometry, build specifications, acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: accelerated occlusion / fouling challenge	UNKNOWN — rig concept finalized, build type, instrumentation	fixed geometry, flow rate, duration, occlusion; acceptance instrumentation
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope	UNKNOWN — rig concept generated, build type, instrumentation	fixed geometry, flow rate, duration, occlusion; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
catheter thrombosis	Fails by catheter thrombosis due to occlusion	Design for fail-safe, i.e. the catheter is designed to be self-cleaning	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
proximal occlusion by tissue ingrowth	Fails by proximal occlusion by tissue ingrowth	Design for fail-safe, i.e. the catheter is designed to be self-cleaning	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
lumen collapse under external pressure	Fails by lumen collapse under external pressure	Design for fail-safe, i.e. the catheter is designed to be self-cleaning	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
encrustation / mineral deposition	Fails by encrustation / mineral deposition	Design for fail-safe, i.e. the catheter is designed to be self-cleaning	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
regime transition to turbulence	Fails by regime transition to turbulence	Design for fail-safe, i.e. the catheter is designed to be self-cleaning	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)
check-valve seat wear	Fails by check-valve seat wear	Design for fail-safe, i.e. the catheter is designed to be self-cleaning	UNKNOWN (no sourced value)	UNKNOWN (no sourced value)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
catheter thrombosis	Fails by catheter thrombosis: the device is designed to not be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)
proximal occlusion by tissue ingrowth	Fails by proximal occlusion by tissue ingrowth: the device is designed to not be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)
lumen collapse under external pressure	Fails by lumen collapse under external pressure: the device is designed to not be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)
encrustation / mineral deposition	Fails by encrustation / mineral deposition: the device is designed to not be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)
regime transition to turbulence	Fails by regime transition to turbulence: the device is designed to not be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)
check-valve seat wear opening	Fails by check-valve seat wear opening: the device is designed to not be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)	Design: catheter is not intended to be used in high flow conditions (e.g. 100 ml/min) and the catheter is not intended to be used in high flow conditions (e.g. 100 ml/min)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct a prospective, double-blind randomized controlled trial comparing the device to standard of care (SOC) in patients with heart failure (HF) and show that the device is superior to SOC in terms of clinical outcomes (e.g. mortality, hospitalization, quality of life) (SOURCE: [1])	Conduct a prospective, double-blind randomized controlled trial comparing the device to standard of care (SOC) in patients with heart failure (HF) and show that the device is superior to SOC in terms of clinical outcomes (e.g. mortality, hospitalization, quality of life) (SOURCE: [1])	Conduct a prospective, double-blind randomized controlled trial comparing the device to standard of care (SOC) in patients with heart failure (HF) and show that the device is superior to SOC in terms of clinical outcomes (e.g. mortality, hospitalization, quality of life) (SOURCE: [1])	Conduct a prospective, double-blind randomized controlled trial comparing the device to standard of care (SOC) in patients with heart failure (HF) and show that the device is superior to SOC in terms of clinical outcomes (e.g. mortality, hospitalization, quality of life) (SOURCE: [1])
VF-002	run killer experiment: falsification test: from kill data (e.g. catheter failure, device failure, patient death) (SOURCE: [2])	run killer experiment: falsification test: from kill data (e.g. catheter failure, device failure, patient death) (SOURCE: [2])	run killer experiment: falsification test: from kill data (e.g. catheter failure, device failure, patient death) (SOURCE: [2])	run killer experiment: falsification test: from kill data (e.g. catheter failure, device failure, patient death) (SOURCE: [2])
VF-003	bench flow-loop characterization across operating range: Acceptance criteria: (1) device flow rate (L/min) is within 5% of target flow rate (L/min) (SOURCE: [3])	bench flow-loop characterization across operating range: Acceptance criteria: (1) device flow rate (L/min) is within 5% of target flow rate (L/min) (SOURCE: [3])	bench flow-loop characterization across operating range: Acceptance criteria: (1) device flow rate (L/min) is within 5% of target flow rate (L/min) (SOURCE: [3])	bench flow-loop characterization across operating range: Acceptance criteria: (1) device flow rate (L/min) is within 5% of target flow rate (L/min) (SOURCE: [3])
VF-004	pressure-drop mapping vs flow (identify pressure drop locations) (SOURCE: [4])	pressure-drop mapping vs flow (identify pressure drop locations) (SOURCE: [4])	pressure-drop mapping vs flow (identify pressure drop locations) (SOURCE: [4])	pressure-drop mapping vs flow (identify pressure drop locations) (SOURCE: [4])
VF-005	delivered-flow trend below baseline in adverse conditions (e.g. high flow, high pressure) (SOURCE: [5])	delivered-flow trend below baseline in adverse conditions (e.g. high flow, high pressure) (SOURCE: [5])	delivered-flow trend below baseline in adverse conditions (e.g. high flow, high pressure) (SOURCE: [5])	delivered-flow trend below baseline in adverse conditions (e.g. high flow, high pressure) (SOURCE: [5])
VF-006	kink/collapse test across duty cycle; flow kink/collapse test across duty cycle (SOURCE: [6])	kink/collapse test across duty cycle; flow kink/collapse test across duty cycle (SOURCE: [6])	kink/collapse test across duty cycle; flow kink/collapse test across duty cycle (SOURCE: [6])	kink/collapse test across duty cycle; flow kink/collapse test across duty cycle (SOURCE: [6])
VF-007	pressure-flow mapping across the full operating range (SOURCE: [7])	pressure-flow mapping across the full operating range (SOURCE: [7])	pressure-flow mapping across the full operating range (SOURCE: [7])	pressure-flow mapping across the full operating range (SOURCE: [7])
VF-008	reverse-leakage rate measurement after accelerated cycling (measure quantity of reverse leakage) (SOURCE: [8])	reverse-leakage rate measurement after accelerated cycling (measure quantity of reverse leakage) (SOURCE: [8])	reverse-leakage rate measurement after accelerated cycling (measure quantity of reverse leakage) (SOURCE: [8])	reverse-leakage rate measurement after accelerated cycling (measure quantity of reverse leakage) (SOURCE: [8])

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	Medical-grade silicone elastomer	Not Established	Not Established	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	Not Established	Not Established	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (Benching) to ensure that the hardware is built correctly and meets the design criteria	physical verification (Benching) to ensure that the hardware is built correctly and meets the design criteria	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope	per method	constraint-compliance envelope	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Conduct a prospective	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	bench flow-loop characterization	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	pressure-drop mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	delivered-flow trend	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	kink/collapse test across	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	pressure-flow mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	reverse-leakage rate	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.